Abtin Aryanpour

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Education

University of Waterloo Waterloo, ON

Candidate for BASc in Computer Engineering, Co-op Program *Expected Apr. 2031*

- Coursework: Fundamentals of Programming (C++), Linear Algebra, Calculus, Mechanics, Engineering Design

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: PyTorch, NumPy, OpenCV, scikit-learn, Django, Streamlit, jQuery
Tools & Platforms: Git, GitHub Actions (CI), pytest, uv, CMake, SQLite, AWS (EC2, CloudFront)

Projects

Andon: Machine-Sound Anomaly Detection | *Python, PyTorch, C, Pico 2* | GitHub Oct. 2026 – Present

- Built an anomaly detector trained only on healthy machine audio: a **98k-weight CNN** reached **92.13% mean AUC** on held-out DCASE 2020 data vs. 79.70% for the reference autoencoder, winning 5 of 6 types (3 seeds)
- Reproduced the published DCASE baseline in PyTorch within **0.88 AUC points** on all 6 machine types, and released checkpoints so two commands re-check the held-out CNN results from a fresh clone
- Turned scores into an andon light with SPC limits (mean + $2\sigma/3\sigma$), hysteresis, a latching stop and a dashboard; on held-out eval clips it caught **65.57% of 30 s faults**, a mean **10.57 s** after onset, at 5.53 false alarms per hour
- Quantized the CNN to int8 for a Cortex-M33 with a custom fixed-point reference: **98.7 KB** (from 383.0 KB) at **92.08% held-out AUC**, with C headers and bit-exact test vectors (export verified in CI); now porting to a Pico 2

Image Stitcher, Waterloo Aerial Robotics | *PyTorch, OpenCV* | PR #215 (in review) Oct. 2026 – Present

- Developed a groundside orthomosaic stitcher on the autonomy subteam, merging out-of-order, GPS-tagged photos from **3 simulated drones** into one georeferenced map using XFeat learned features and RANSAC
- Fused images with GPS to recover heading, aligning photos to **4.3–6.9 cm** on average at 3 m simulated GPS error
- Designed a bundle-adjustment-style least-squares solver, cutting simulated misalignment from **13–41 to 4–7 cm**
- Handled 180-degree turns and repetitive terrain, forming one map on **10/10** simulated 91-photo missions
- Added lens undistortion and blur, exposure and tilt checks; built a ground-truth mission simulator and **80 tests**

CyberShield: Full-Stack Discussion Forum | *Python, Django, JavaScript, Tailwind CSS* | GitHub Mar. 2025

- Sole developer of a responsive forum where users post, comment and vote, with sorting, topic filtering and fuzzy search (Fuse.js), built on Django, jQuery (AJAX) and SQLite
- Secured accounts with Django authentication, CSRF protection and reCAPTCHA on login and signup; added rich-text posts (CKEditor) and profile picture uploads
- Deployed live behind AWS CloudFront with a seeded demo dataset; won **1st place** in Coding & Programming at the Canadian National Leadership Conference 2025 (32 teams)

Experience

Autonomy Software Member Sep. 2026 – Present

Waterloo Aerial Robotics Group *Waterloo, ON*

- Write Python for the autonomy subteam's shared codebase, submitting all changes as reviewed pull requests
- Built the team's groundside image stitcher (PR #215) and worked through two rounds of review changes (lens undistortion, photo quality checks, numbered map saves), including a fix for sensor noise fooling the blur check

Treasurer Sep. 2025 – Jun. 2026

Bayview Secondary School *Richmond Hill, ON*

- Managed a **$21,000** student budget across **42 clubs** serving 2,000 students
- Reviewed and approved each club's budget, calculating allocations and negotiating changes in meetings with club presidents and school administrators

Homework Helper & Assistant Teacher Sep. 2022 – Jul. 2025

Spirit of Math Schools Inc. *Richmond Hill, ON*

- **Homework Helper (2024–2025):** Ran a drop-in homework help classroom as one of two staff, triaging questions from students ranging from kindergarten to Grade 11 and diagnosing where each one was stuck
- **Assistant Teacher (2022–2024):** Supported classes of 16 students, teaching lessons and contest-style algebra